OPTIMIZATION 1 ⇒ [Need for calculus in ML ⇒]

(1) Agenda ⇒

1. Classifier
2. SEARCHING ALGO ⇒ BRUTE FORCE ✗
3. Optimization PROBLEM
4. FUNCTION
5. LIMITS ⇒ CONTINUITY & Differentiability

(*) Classifier ⇒

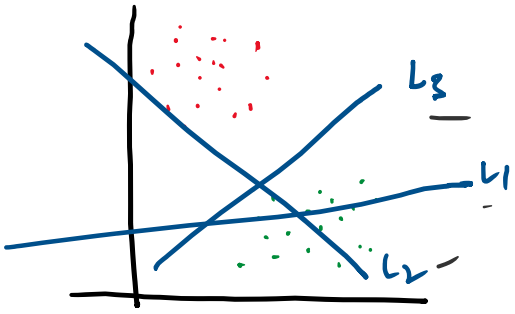
(A) Loan Approval PROBLEM

= Bank Approver

→ Approved
→ Rejected

[CS → financial TRUSTWORTHINESS
HDFC Bank →

JOB → Approve OR Reject



→ infinite possible lines

⇒ Best one ⇒ (L3) ⇒

quantifying ⇒ max. avg. distance of points from the line

ERROR func^n

all possible times ⇒ we will find ERROR value

(*) Loss func^n | ERROR func^n

⇒

Loss func^n ⇒

$L(\text{Dataset}, w_1, w_0) =$ o/p value which would be a real no.

w_1 : weight, w_0 : intercept

→ max. | min.? ⇒ minimize loss

⇒ Loss func^n will help quantify how good or bad line is.

① ERROR ⇒ ↑

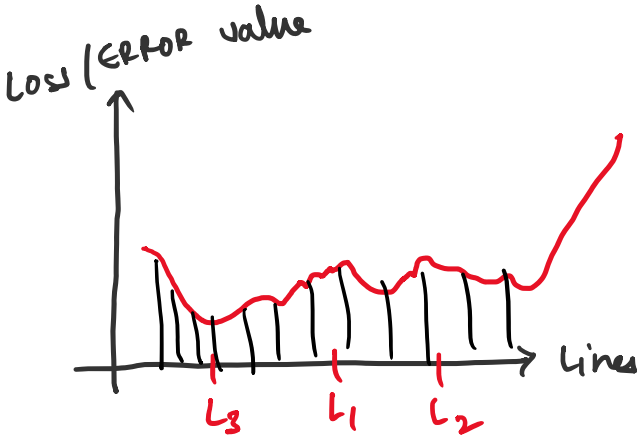
(*) Loss | ERROR ↓

(*) Maximize Gain = Minimise Loss

② Gain function → how good classifier is

Gain function → how good classifier is

Loss funⁿ → how Bad your classifier is



L3 ⇒ ✓ giving minimum Loss

funcⁿ ⇒ (Data, w, w0) → ℝ

Gradient Descent ⇒ ⇒ Loss ↓ ↓

Simple SEARCHING ALGORITHM → GD ⇒ GPT, Transformer, LLM, LR, Logistic Regⁿ

infinite no. of lines

Line → that minimizes ERROR / Loss → Best Line

⇒ [Linear Search] ✗

⇒ Best value of w & w0

2 features ⇒ $\vec{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} + w_0$

CS ✓ y

4 DEBTS

how many elements

Params to learn ⇒ 3 parameters

3 features ⇒ $[\,]_3 + w_0 = 4$ params

⇒ (d) ⇒ (d+1) parameters To learn

$w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$ w_0

→ Best value of w_1 w_2 & w_0

Best eqⁿ $[\, w_1^* x_1 + w_2^* x_2 + w_0^* = 0 \,]$

How do I find Best Ws ⇒

Linear SEARCH ⇒ w & w_0

=

	L_1	L_2	L_3
w_1 → [-10, 10]	-10 →	-9.9 →	-9.8 →
w_2 → [-10, 10]	-10	-10	-10

Support

(*) Support

(*) $w_2 \rightarrow [-10, 10]$ -10 -10 -10

$w_0 \rightarrow [-10, 10]$ -10 -10 -10

$L_1 \rightarrow$ Loss (store) $L_2 \rightarrow$ Loss (store)

⇒ Try each value & see where does loss Best w exists ?

store the loss value for L_1

(*) Loss < previous one

⇒ Better line

⇒ check with other values as well

$w_1 \rightarrow$ fixed, $w_2 \rightarrow -10 \rightarrow -9.9 \rightarrow -9.8$

$w_0 \rightarrow$ fixed

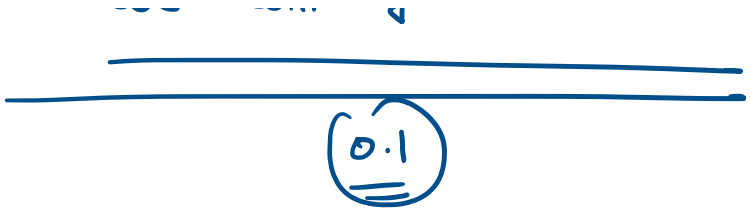
Loss ↓ storing

w_1 - fixed, w_2 - fixed

$w_0 \rightarrow -10 \rightarrow -9.9 \rightarrow -9.8 \ldots$

$[-10, +10]$ $-10 \rightarrow +10$

(*) we will go till +10



Step size ⇒

② Trying all BRUTE FORCE combinations of w_1 w_2 & w_0

⇒ [BAD METHOD]

⇒ Total possible values ⇒?

w_1 [-10, 10]
w_2 [-10, 10]
w_2 [-10, -10]

⇒ Combinations possible

Step size = 0.1

[10 → -9.9 → -9.8 ---- +10]

[-10 ---- -0.1]
—— 100 values ——

[0] ⇒ (201) values

[0.1 ---- +10]
— 100 values

↳ w_1

⇒ Trying for w_2 & w_3

⇒ Trying for w_2 & w_3

⇒ $(201)^3$ comb^{n}s = To try

$(201)^3$ ⇒ Step size = 0.1

↑

0.05

0.01

⇒ ↑ Big comp ⇒ (Brute line ⇒

Linear search → Not a good idea

Laptop ⇒ 10^6 operations / second

⇒ $(201)^3$ comb^{n}s = 8 seconds

⇒ $\frac{(201)^3}{10^6}$ ⇒ $\frac{(201)^3 \times 10^{-6}}{11}$

8 seconds

2 features → 3 params

4 features → 4 params ⇒ $(201)^4$

① 3 features → 4 params ⇒ $(201)^4$

$(201)^4 \times 10^{-6} =$ 27 mins

$\frac{1}{60}$ =

② 4 features → 5 params ⇒

Expensive

Not possible to try every single value

=

Time ↑ exponentially

linear time

calculus ⇒

using

very Bad idea

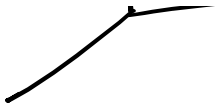
= 4 params

In a three-dimensional space, how many parameters or variables are involved in defining a 3D hyperplane?

From <https://www.scaler.com/instructor/meetings/i/optimization-1-the-need-for-calculus-in-ml-85/>



⇒



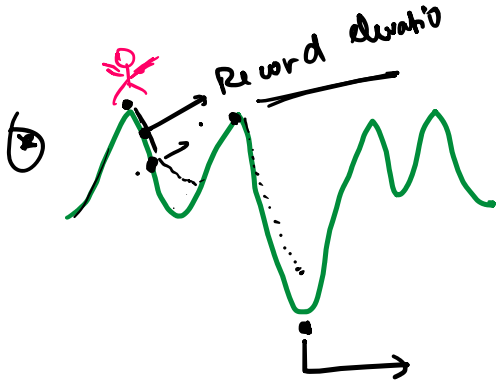
we will use optimization & calculus

→ solving optimization problem

↳ any problem statement where you need to find best value of variables or params

① Gradient Descent (**)

(a)



↳ Minima

(*) Lowest valley

(*) Brute force approach

→ Rewarding elevation

↳ very long

↳ lifetime

⇒ seeing the ground & moving

⇒ Gradient Descent

(*) Linear line

Brute force

↳ Min, Hours

$$f(x) = (x-5)^2$$

$=$

find such value of x so that entire func^n is minimum

(*)

$\rightarrow x \Rightarrow$ min. possible value

at $x=5 \Rightarrow f(x) = 0$ ✓

$\Rightarrow$ Best value

$\Rightarrow$ $\underset{x}{\text{argmin}} \; (x-5)^2$

$x \rightarrow 0$

Best value of x

optimization PROBLEM ✓

$\Rightarrow$ optimizing $\Rightarrow$ (x)

$(x-5)^2$

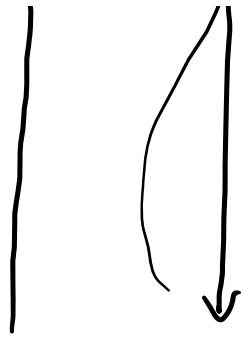
a (max. ↑ / min) $(x-5)^2$ ↓ min

$\Rightarrow$ Gradient Descent $\Rightarrow$

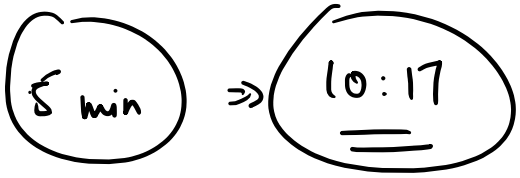
↳ maxima / minima

↳ calculus in multivariate

↳ single variate



↳ SINGLE VARIATE
↳ DERIVATIVE & SLOP
↳ LIMIT, CONTINUITY THM ⇒



② PROBLEM → CLASSIFICATION PROBLEM ⇒

$$D = \{ (x_i, y_i)_{i=1}^{N} \mid x^i \in R^d, \; y^i \in \{-1, +1\} \}$$

Labels · d dimen · Features · $x^1, x^2 \dots x^d$

Classification Labelled DATA PROBLEM

Loan Approval

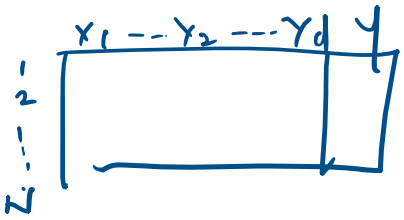
any data for classification would be represented using this eqn

	X_1	X_2	X_3	$\dots$	X_d	Y (Labels)
1						= 1 → Rejected
2						+ 1 → Approved
3						− 1
4						+
⋮						
n						

⇒ DATA will always be given

$X_1, X_2 \dots X_d$ Y → ACTUAL LABEL

$$D = \{(x^i, y^i)\}_{i=1}^{N} \;,\; x^i \in \mathbb{R}^d \;,\; y^i \in \{-1, +1\}$$



n Approach

DATA → given

① OUR goal → find a funⁿ $f(x)$ such that

$f(x)$ = Predicted values $= w^T x + w_0$

② y^i → actual label / output

↳ label for i^{th} DATA point

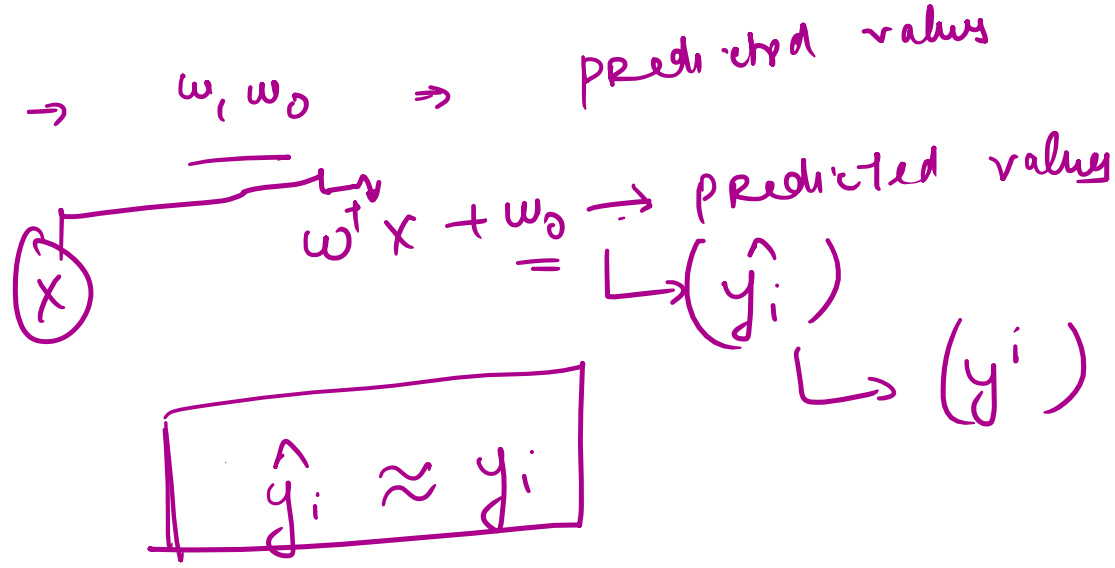
③ $\hat{y}^{(i)}$ → Prediction

↳ Predicted label by your ML model for i^{th} DP

Actual Prediction

$\Rightarrow$ $y^i \approx \hat{y}^{(i)}$?

Approximately



gain funⁿ ⇒

$G(D, w, w_0)$ ⇒ will give how good or Bad a line is

↳ how good classifier performing

(8) w, w_0 ⇒ ✓ max value of gain funⁿ

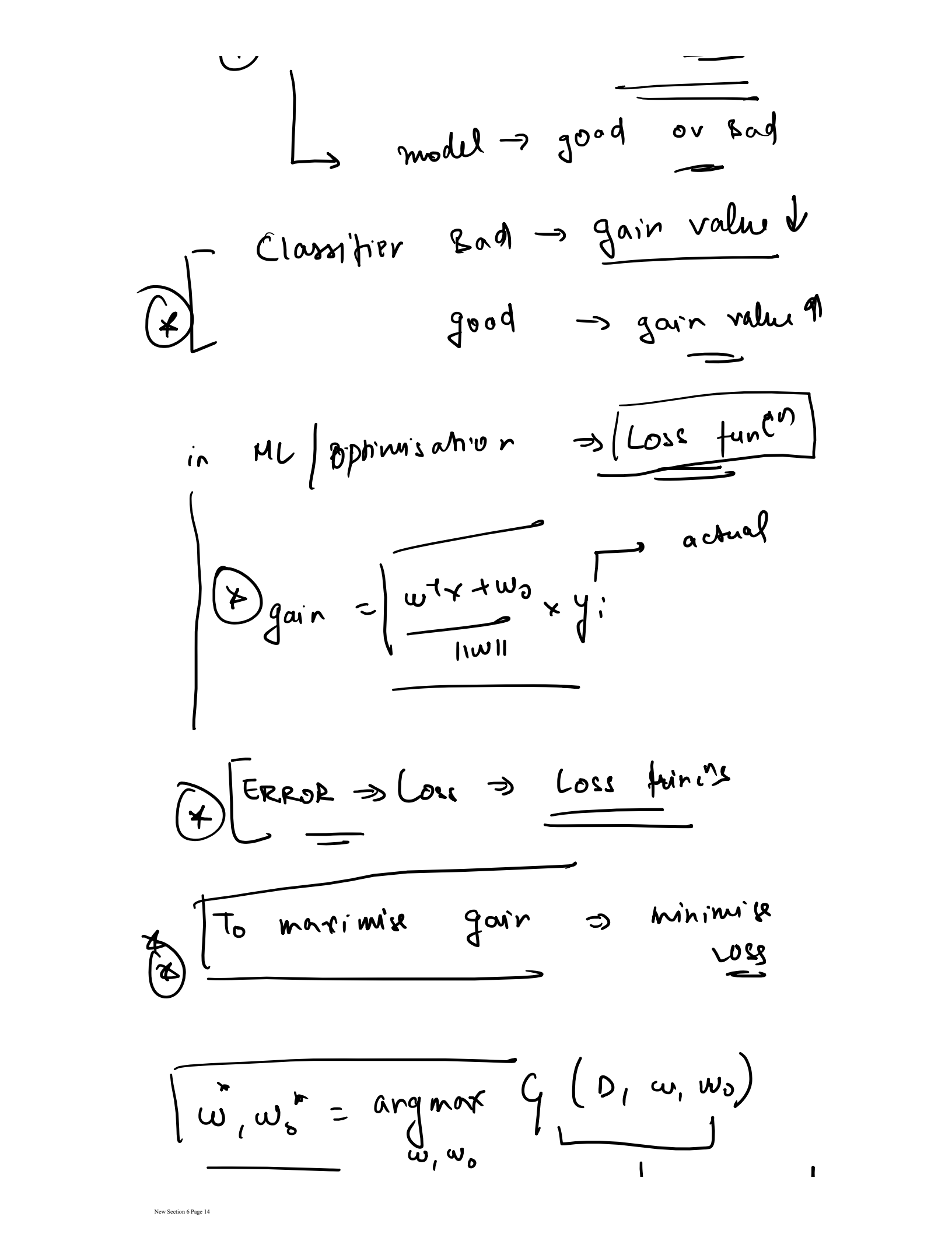
or

✓ min value of loss funⁿ

$\boxed{\text{Gain}(D, w, w_0)}$ → This is the funⁿ i want to optimize

(+)

|


model → good or bad
Classifier Bad → gain value ↓
good → gain value ↑
in ML / optimisation → Loss func^n
gain = (w^T x + w_0 / ||w||) × y_i
actual
ERROR → Loss → Loss func^ns
To maximise gain → minimise loss
w*, w_0* = arg max_{w, w_0} G (D, w, w_0)

w, w_0 — $\max_{w, w_0}$ — ↓ maximized

(1) $g(D, w, w_0) = \frac{1}{n} \sum_{i=1}^{n} \frac{w^T x_i + w_0}{\|w\|} \cdot y_i$ (summation)

(2) $g(D, w, w_0) = \frac{1}{n} \prod_{i=1}^{n} \frac{w^T x_i + w_0}{\|w\|} \cdot y_i$ (product)

=

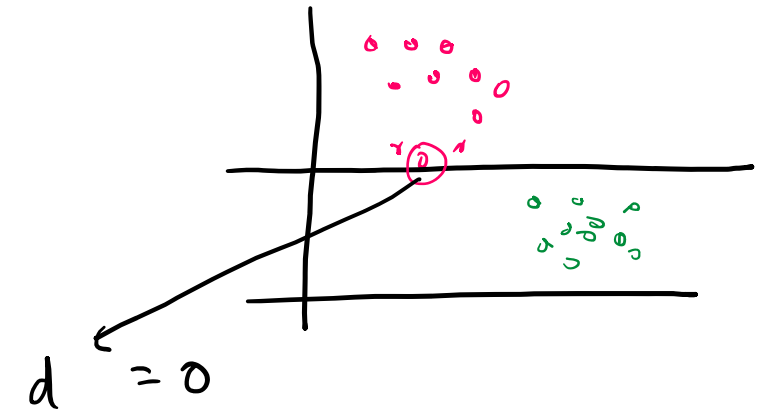
Sum = product → scalar

(2) nd why not 2nd

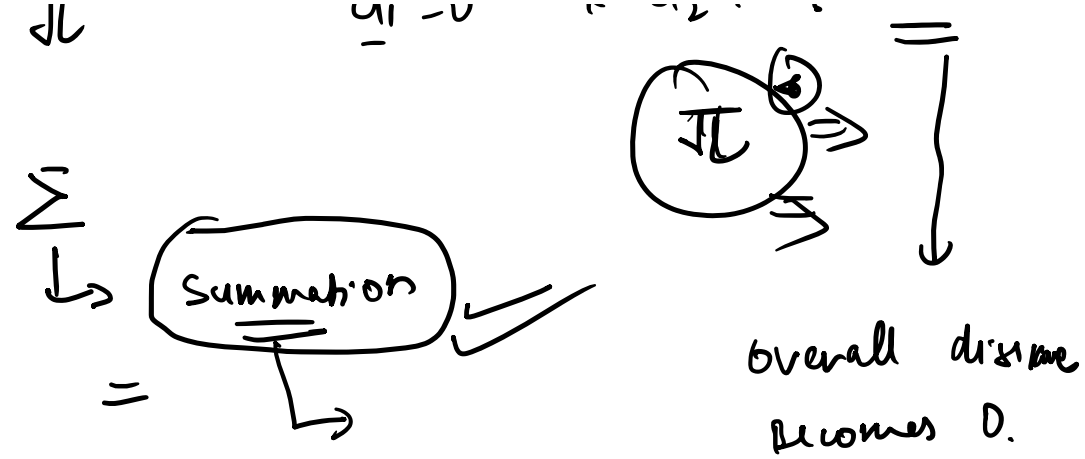
(*) why not multiplication

=

⇒ what if DATA POINT is ON THE Line



$\prod$ $d_1 = 0$ $\times$ $d_2 * d_2 = 0$ = 1



X → optimise for product

Generic flow for any ML task →

1. you are given a labelled data
2. define a model → $f(x) = w^T x + w_0$
3. Loss fun^n
4. optimization → w, w_0

gain/loss → can be changed w.r.t. diff. PROBLEM statement

Statistic

Fraud → loss

Revenue ops → loss

MLE, HUBER LOSS

Loss func^n → optimization Best value

① Functions ⇒

$f(x) = x^2$ → output

- f → name of func^n
- x → input

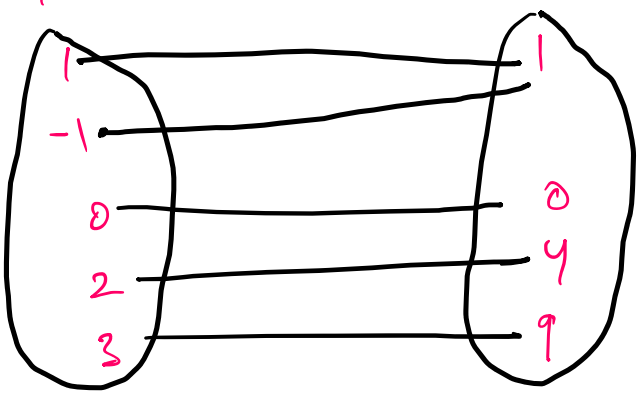
or → $y = x + 5$

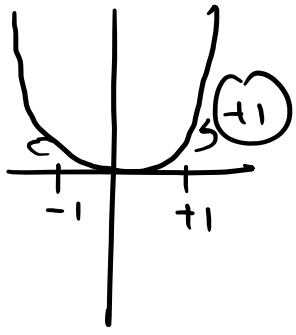
$f(x) = x + 5$

output

each i/p ⇒ exactly 1 output

② i/p → $f(x) = x^2$ → o/p





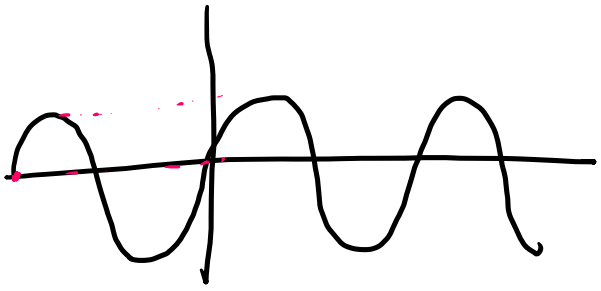
valid func^n

all set of possible inputs

all set of possible output from func^n

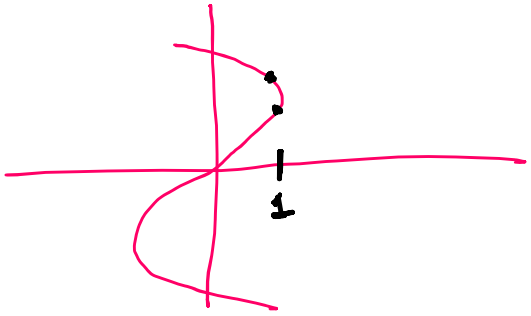
inputs → Domain

turn'n → Range



⇒ valid or invalid

⇒ Sin curve



valid ?

invalid ⇒ ✗

func defⁿ ⇒ for a single value of x

it should have single value of y.

(4)

$x^2 + y^2 = R^2$ ⇒

valid ?

(x=0) → $y^2 = r^2$, $y = \pm r$

x = 2 values of R

invalid

✓ ✗

(1,1)

$\log(0)$ ⇒ undefined

$\log(x)$

- $f(x) = \log(x)$

(*) Domain & Range

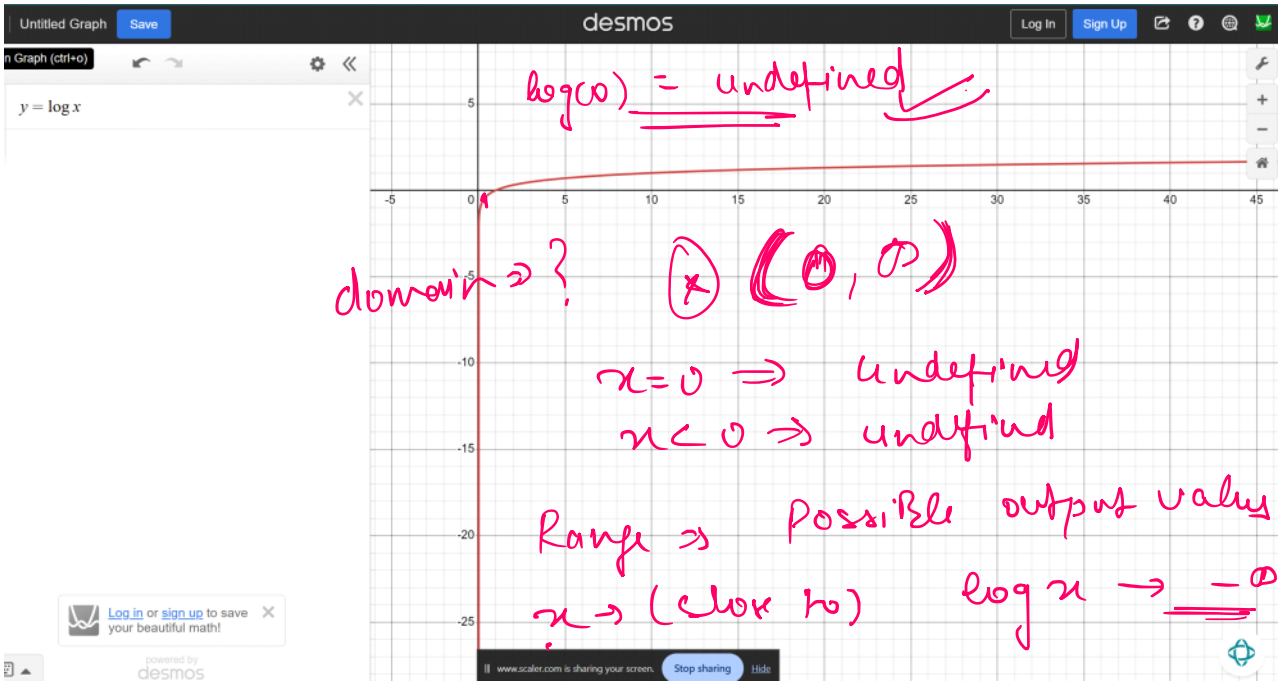
$(0, \infty)$ $(-\infty, \infty)$

([($\log x \rightarrow$ monotonically ↑

(*) $(a,b) \Rightarrow$ a & b are excluded
$[a,b] \Rightarrow$ a & b included
$(a,b] \Rightarrow$ a exclude b included

(*) Limit (*) ND $\Rightarrow$ Next session (*)

$\log x \Rightarrow$ Domain =



$\log(0)$ = undefined

domain → ? (x) $(0, \infty)$

$x = 0 \Rightarrow$ undefined

$x < 0 \Rightarrow$ undefined

Range → possible output values

$x \to$ (close to 0) $\log x \to -\infty$

$x \uparrow \Rightarrow \log x \to +\infty$

$\log(1) = 0$

$\log(0.1) = -1$

$\log(100) = 2$ (100 ↑ 1000)

[Range → $(-\infty, \infty)$]

$x \leq 0$

↓ undefined

questions ?